Computational PDEs — Day 1 classwork

Exercises (15–20 minutes)

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Format

Two questions. Work in pairs if you like.
Use `01_ode_pinn.ipynb` for Question 2 (kernel `comp_pde`).
Write answers in a note or a Markdown cell.

1. **Residual and exact solution** (about 7–8 min) — pen and paper.
2. **Relative L^2 and the BC weight** (about 8–10 min) — short notebook run.

1 Question 1 — Residual and exact solution

Consider

$$u'(x) + 2u(x) = 0, \qquad u(0) = 1, \qquad x \in (0, 1). \tag{1}$$

1. Write the **PDE residual** $r(x)$ that a PINN would drive to zero (same idea as today's $u' + u$).
2. Write the **boundary residual** at $x = 0$.
3. Give the **exact** solution $u(x)$.

Check

If your u is correct, then $r \equiv 0$ and the boundary residual is 0. No training is required for this question.

Your answers (Q1):

$r(x) =$ _____

BC residual = _____

$u(x) =$ _____

2 Question 2 — Relative L^2 and `W_BC`

Open `work/pinn_learning/01_ode_pinn.ipynb` (problem $u' + u = 0$, $u(0) = 1$, exact $u = e^{-x}$). The notebook default is `W_BC` $= 100$.

1. Run once with the default. Record the printed **relative** L^2 against e^{-x}.
2. Set `W_BC` $= \mathbf{1}$, re-run training from a fresh model (re-run from the model-creation cell), and record the error.
3. Set `W_BC` $= \mathbf{10}$, same procedure, and record the error.
4. In one or two sentences: which weight looked best, and why a tiny PDE loss alone is not enough to trust the answer.

Reminder

Relative L^2: $\|u_\theta - u_{\text{ex}}\|_2 / \big(\|u_{\text{ex}}\|_2 + \varepsilon\big)$ (`relative_l2`). Keep the default `EPOCHS`; only change `W_BC`.

What to hand in (today)

- Q1: the three short formulae (residual, BC residual, exact u).
- Q2: the table below, plus the short answer to Q2(4).

`W_BC`	relative L^2	note (optional)
100 (default)		
1		
10		

Q2(4) answer:

Time box

Aim to finish both questions within 20 minutes. If the machine is slow, reduce `EPOCHS` to 2000 for all three runs and say so in the note column.